

Student registration form:

Frontend: React

Backend: Express (Node.js)

Database: MongoDB

Let's go step-by-step so you can build it clearly.

 Project Overview

Goal: Create a Student Registration System where users can:

Add new students (name, email, course, contact, etc.)

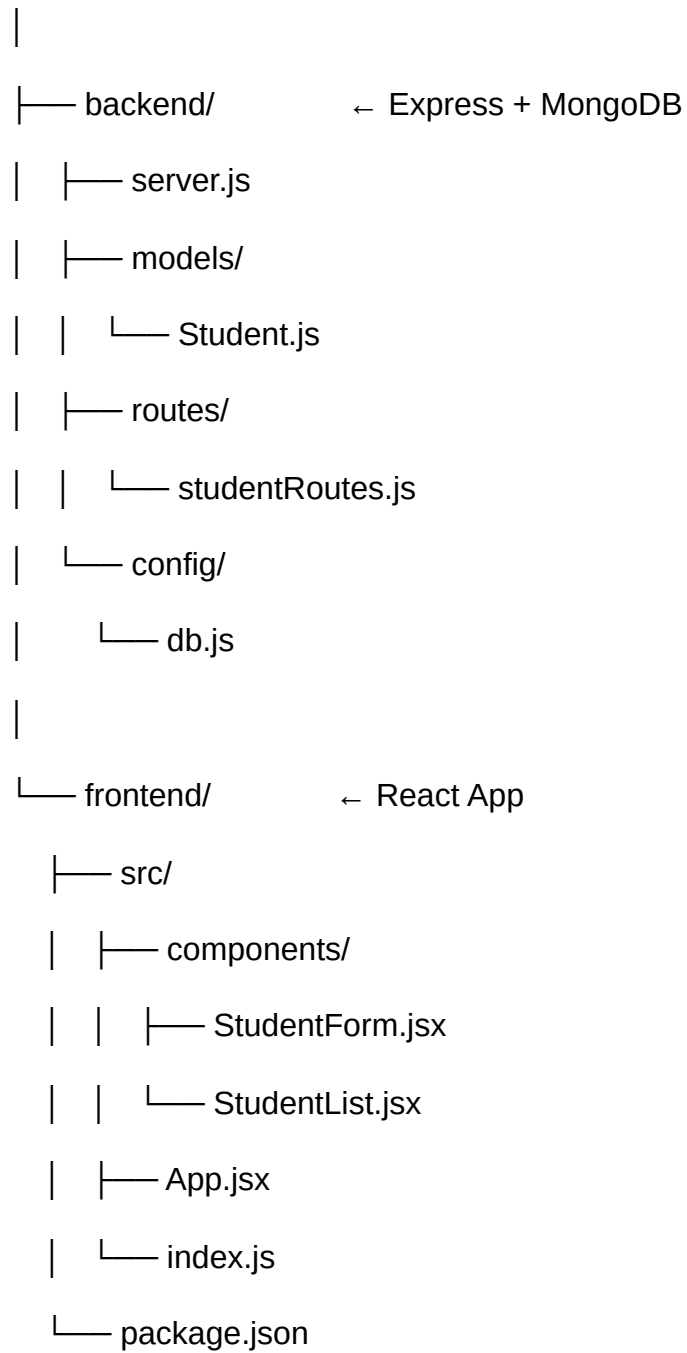
View all registered students

Update student details

Delete a student

🔗 Project Structure

student-registration/



1. Backend Setup (Express + MongoDB)

Step 1: Initialize Backend

```
mkdir backend
```

```
cd backend
```

```
npm init -y
```

```
npm install express mongoose cors dotenv
```

Step 2: Create server.js

```
import express from "express";
```

```
import dotenv from "dotenv";
```

```
import cors from "cors";
```

```
import connectDB from "../config/db.js";
```

```
import studentRoutes from "../routes/studentRoutes.js";
```

```
dotenv.config();
```

```
const app = express();
```

```
// Middleware
```

```
app.use(cors());
```

```
app.use(express.json());
```

```
// Routes
```

```
app.use("/api/students", studentRoutes);
```

```
// Connect to MongoDB and start server

connectDB();

const PORT = process.env.PORT || 5000;

app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
```

Step 3: config/db.js

```
import mongoose from "mongoose";

const connectDB = async () => {
  try {
    await mongoose.connect(process.env.MONGO_URI);
    console.log("MongoDB Connected");
  } catch (err) {
    console.error(err.message);
    process.exit(1);
  }
};

export default connectDB;
```

Step 4: Create models/Student.js

```
import mongoose from "mongoose";

const studentSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true },
  course: { type: String, required: true },
  phone: { type: String },
});

const Student = mongoose.model("Student", studentSchema);
export default Student;

---
```

Step 5: Create routes/studentRoutes.js

```
import express from "express";
import Student from "../models/Student.js";

const router = express.Router();

// Create student
router.post("/", async (req, res) => {
```

```
try {  
  const student = new Student(req.body);  
  await student.save();  
  res.json(student);  
} catch (err) {  
  res.status(400).json({ message: err.message });  
}  
});
```

// Get all students

```
router.get("/", async (req, res) => {  
  const students = await Student.find();  
  res.json(students);  
});
```

// Update student

```
router.put("/:id", async (req, res) => {  
  const student = await Student.findByIdAndUpdate(req.params.id, req.body, { new: true });  
  res.json(student);  
});
```

// Delete student

```
router.delete("/:id", async (req, res) => {  
  await Student.findByIdAndDelete(req.params.id);  
  res.json({ message: "Student deleted" });  
});
```

```
export default router;
```

2. Frontend Setup (React)

Step 1: Create React App

```
cd ..
```

```
npx create-react-app frontend
```

```
cd frontend
```

```
npm install axios
```

Step 2: src/App.jsx

```
import React from "react";
```

```
import StudentForm from "../components/StudentForm";
```

```
import StudentList from "../components/StudentList";
```

```
function App() {
```

```
  return (
```

```
    <div className="container">
```

```
      <h1><img alt="graduation cap icon" data-bbox="125 918 145 938"/> Student Registration</h1>
```

```
    <StudentForm />

    <StudentList />

  </div>

);

}
```

```
export default App;
```

```
---
```

Step 3: src/components/StudentForm.jsx

```
import React, { useState } from "react";

import axios from "axios";

const StudentForm = () => {

  const [student, setStudent] = useState({ name: "", email: "", course: "", phone: "" });

  const handleChange = (e) => setStudent({ ...student, [e.target.name]: e.target.value });

  const handleSubmit = async (e) => {

    e.preventDefault();

    await axios.post("http://localhost:5000/api/students", student);

    alert("Student Registered!");

    setStudent({ name: "", email: "", course: "", phone: "" });

  };

};
```



```
return (  
  <form onSubmit={handleSubmit}>  
    <input name="name" value={student.name} onChange={handleChange} placeholder="Name"  
required />  
    <input name="email" value={student.email} onChange={handleChange} placeholder="Email"  
required />  
    <input name="course" value={student.course} onChange={handleChange} placeholder="Course"  
required />  
    <input name="phone" value={student.phone} onChange={handleChange} placeholder="Phone" />  
    <button type="submit">Register</button>  
  </form>  
);  
};  
  
export default StudentForm;
```

Step 4: src/components/StudentList.jsx

```
import React, { useEffect, useState } from "react";  
import axios from "axios";  
  
const StudentList = () => {  
  const [students, setStudents] = useState([]);
```

```
const getStudents = async () => {  
  const res = await axios.get("http://localhost:5000/api/students");  
  setStudents(res.data);  
};
```

```
useEffect(() => { getStudents(); }, []);
```

```
return (  
  <div>  
    <h2>Registered Students</h2>  
    <ul>  
      {students.map((s) => (  
        <li key={s._id}>  
          {s.name} ({s.course}) — {s.email}  
        </li>  
      ))}  
    </ul>  
  </div>  
);  
};
```

```
export default StudentList;
```

3. Environment File (.env)

In your backend folder:

MONGO_URI=mongodb://127.0.0.1:27017/studentDB

PORT=5000

4. Run the Project

In two terminals:

Terminal 1 - Backend

cd backend

node server.js

Terminal 2 - Frontend

cd frontend

npm start

 Output

You'll get a web app where you can:

Add students through the form.

View all students below.

(Optionally) implement Edit/Delete later.